Definition of a Cut in Graph Theory

1 Definition: Cut (in a Graph)

Let $G = (V, E)$ be an undirected graph, where V is the set of vertices and E is the set of edges.

A **cut** is a partition of the vertex set V into two disjoint subsets, usually denoted S and $V \setminus S$ (or S and T).

We say that an edge $e = \{u, v\} \in E$ **crosses the cut** (or is a **cut edge**) if its endpoints belong to different subsets. That is, the edge crosses the cut if $u \in S$ and $v \in V \setminus S$, or vice versa.

The Size of a Cut

The size of a cut, denoted $|C|$ or $\delta(S)$, is the number of edges crossing the cut:

$$\delta(S) = |\{\{u, v\} \in E : u \in S \text{ and } v \notin S\}|$$

(Note: In a directed graph, a cut is defined similarly, but we only count edges that go from S to $V \setminus S$.)

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2 Geometric & Algebraic Intuition

- **Geometrically:** Imagine the graph drawn on a piece of paper. A cut is a literal line (or curve) drawn through the graph that divides the vertices into two groups. The "size" of the cut is the number of edges that you physically have to cut with scissors to separate the two groups.
- **Algebraically (using a coloring vector):** If we assign every vertex to a set using a function $\chi : V \to \{-1, +1\}$, an edge $\{u, v\}$ crosses the cut if and only if $\chi(u) \neq \chi(v)$. The size of the cut is the sum over all edges of the indicator that they cross:

$$|C| = \sum_{\{u,v\} \in E} \frac{1 - \chi(u)\chi(v)}{2}$$

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3 Examples

1. **Complete Graph (K_n):** If you have n vertices all connected to each other, and you split them into sets of size k and $n - k$, every vertex in the first set is connected to every vertex in the second set. The size of this cut is exactly $k(n - k)$. The **Max-Cut** of a complete graph is $\lfloor n^2/4 \rfloor$ (by splitting the vertices as evenly as possible).

2. **Bipartite Graph:** A graph is bipartite if and only if its vertices can be partitioned into two sets S and T such that *every* edge in the graph crosses the cut. The size of this cut is exactly $|E|$.

3. **Cycle Graph (C_n):** If you have a ring of n vertices, cutting it into two contiguous halves will cut exactly 2 edges.

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4 Related Concepts

- **Min-Cut:** The partition that minimizes the number of crossing edges. (Famous for the **Max-Flow Min-Cut Theorem** in network flow).
- **Max-Cut:** The partition that maximizes the number of crossing edges. (Famous because it is NP-hard, as discussed earlier).
- **Sparsest Cut:** A cut that minimizes the ratio of crossing edges to the size of the smallest resulting set. (Highly relevant in graph clustering and machine learning).
- **Cut Vertex / Articulation Point:** A single vertex whose removal disconnects the graph (different from a cut edge, which is an edge whose removal disconnects the graph).